

# Lab: Markdown and Reproducible Reports

Actuarial Data Science - Open Learning Resource

Fei Huang, UNSW Sydney

## Learning Objectives

- Learn how to use Markdown and reproducible reporting tools to write academic reports
- Understand the differences between R Markdown, Quarto, and other markdown-based reporting frameworks
- Practice creating reproducible documents with embedded code in both R and Python

## Markdown and Reproducible Reports Introduction

This is a Markdown document that can be used to create **HTML**, **PDF**, and **MS Word** documents with embedded code. Markdown is a simple formatting syntax that is widely used for creating reproducible reports. This lab introduces various markdown-based reporting frameworks, including R Markdown, Quarto, and Jupyter notebooks.

## Why Use Markdown for Reproducible Reports?

There are many advantages to using Markdown-based reporting in your work:

- **Reproducibility:** Code and outputs are directly embedded in the document, ensuring results can be reproduced
- **Automation:** When data or code changes, all outputs (figures, tables, numbers) are automatically updated
- **Version Control:** Markdown files work well with Git and other version control systems
- **Multiple Output Formats:** Create HTML, PDF, Word, and presentation slides from the same source
- **Collaboration:** Easy to share and collaborate on documents with embedded code
- **Cross-Platform:** Works on Windows, Mac, and Linux

## Available Frameworks

### R Markdown

- **Primary Language:** R
- **Package:** rmarkdown
- **Strengths:** Excellent R integration, mature ecosystem, extensive customisation
- **Best For:** R-focused projects, statistical analysis, and academic publishing

## Quarto

- **Primary Languages:** R, Python, Julia, Observable JavaScript
- **Package:** quarto
- **Strengths:** Multi-language support, modern features, and excellent documentation
- **Best For:** Multi-language projects, modern workflows, and cross-language collaboration

## Jupyter Notebooks

- **Primary Languages:** Python, R, Julia, and many others
- **Format:** .ipynb files
- **Strengths:** Interactive computing, rich output, and extensive language support
- **Best For:** Data science, machine learning, and interactive analysis

## Other Options

- **Bookdown:** An extension of R Markdown for books and long documents
- **Distill:** An R Markdown extension for scientific and technical writing
- **Hugo:** A static site generator with Markdown support

In this course, we will primarily use Quarto (.qmd).

## Installation

### For R Markdown

Before installing R Markdown, you need to have R and RStudio installed:

1. **Install R:** Go to [R Project](#) and choose the correct version for your system
2. **Install RStudio:** Go to [RStudio](#) and install the appropriate version
3. **Install rmarkdown:** Open RStudio and run:

```
# install.packages("rmarkdown")
```

### For Quarto

Quarto is the newest and most versatile option:

1. **Install Quarto:** Go to [Quarto](#) and download the installer
2. **Install R:** If using R with Quarto, install R as above
3. **Install Python:** If using Python with Quarto, install Python from [python.org](#)

## For Jupyter Notebooks

1. **Install Python:** Download from [python.org](https://python.org)
2. **Install Jupyter:** Open terminal/command prompt and run:

```
pip install jupyter
```

3. **Install R kernel (optional):** If you want to use R in Jupyter:

```
install.packages("IRkernel")  
IRkernel::installspec()
```

*Note: For PDF output, you will need a LaTeX distribution. We recommend TinyTeX:*

```
```{r}  
#| echo: true  
  
# To install TinyTeX (works with R Markdown and Quarto)  
tinytex::install_tinytex()  
  
# To uninstall TinyTeX  
tinytex::uninstall_tinytex()  
```
```

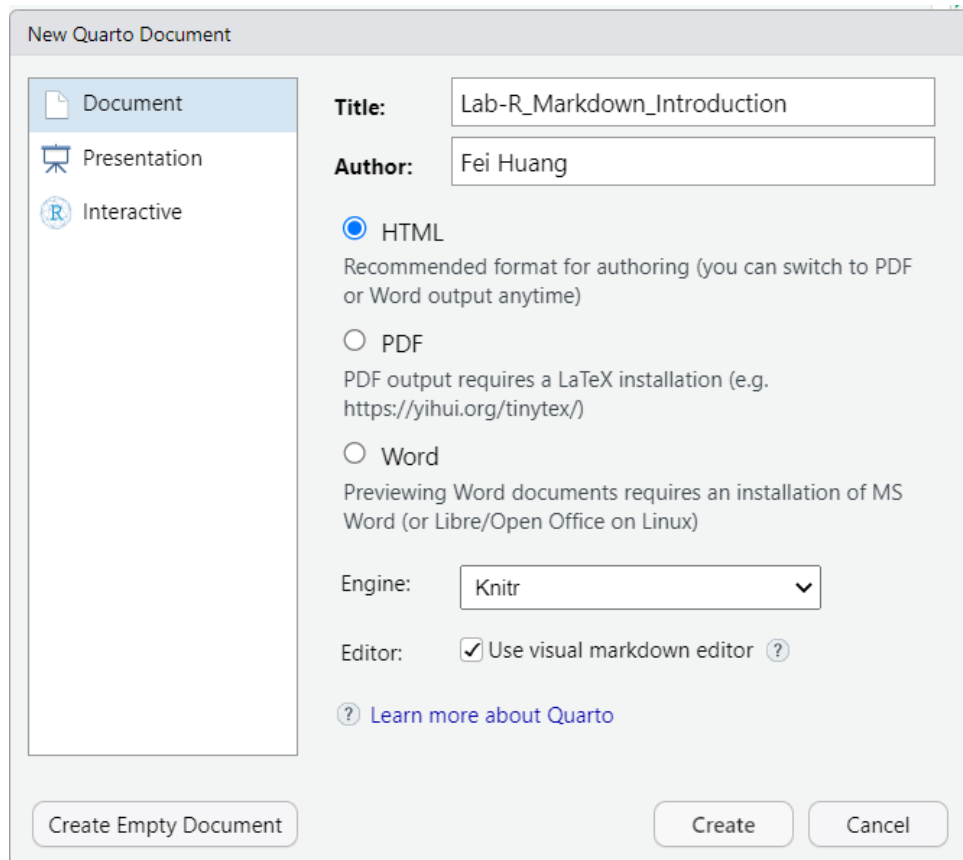
## Creating Your First Document

### R Markdown

In RStudio, click “File” → “New File” → “R Markdown”:

### Quarto

In RStudio, click “File” → “New File” → “Quarto Document”, or create a new `.qmd` file manually.



Add the following YAML header:

```
---  
title: "My Report"  
format: html  
---
```

## Jupyter

Open a terminal/command prompt and run:

```
jupyter notebook
```

### **i** Note

From this point onwards, we will primarily focus on Quarto (.qmd) for reproducible reporting.

## Document Structure

### YAML Header

The YAML header controls document metadata and output format:

```

---
title: "My Amazing Report"
author: "Your Name"
date: today
format:
  html: default
  pdf: default
bibliography: references.bib
---

```

## Markdown Content

Markdown provides simple formatting:

- **Bold** and *italic* text
- Lists and numbered lists
- Headers with #, ##, ###
- Links: [GitHub](#)
- Images: ![Caption](image.png)

## Code Chunks

Embed executable code in your document:

### R in Quarto:

```

```{r}
# Your R code here
```

```

### Python in Quarto:

```

```{python}
# Your Python code here
```

```

## Code Chunk Options

Control how code chunks behave (specified using #| in Quarto):

- echo: false — Hide code, show results
- results: "hide" — Show code, hide results
- include: false — Hide both code and results
- warning: false — Suppress warnings
- message: false — Suppress messages
- fig-width: 6, fig-height: 4 — Control figure size

## Example with Options

```
set.seed(123)
x <- rnorm(100)
y <- 2 * x + rnorm(100)
cor(x, y)
```

### Note

The result is hidden due to results: "hide", which looks like the following:

```
```{r}
#| label: simulate-data
#| echo: true
#| results: "hide"
#| warning: false

set.seed(123)
x <- rnorm(100)
y <- 2 * x + rnorm(100)
cor(x, y)
```
```

```
import numpy as np

np.random.seed(123)
x = np.random.normal(0, 1, 100)
y = 2*x + np.random.normal(0, 1, 100)
print(f"Correlation: {np.corrcoef(x, y)[0,1]:.3f}")
```

Correlation: 0.918

### Python Setup

If you encounter errors such as “No module named ...”, you may need to install the required Python packages.

You can install them using:

```
pip install package_name
```

For example:

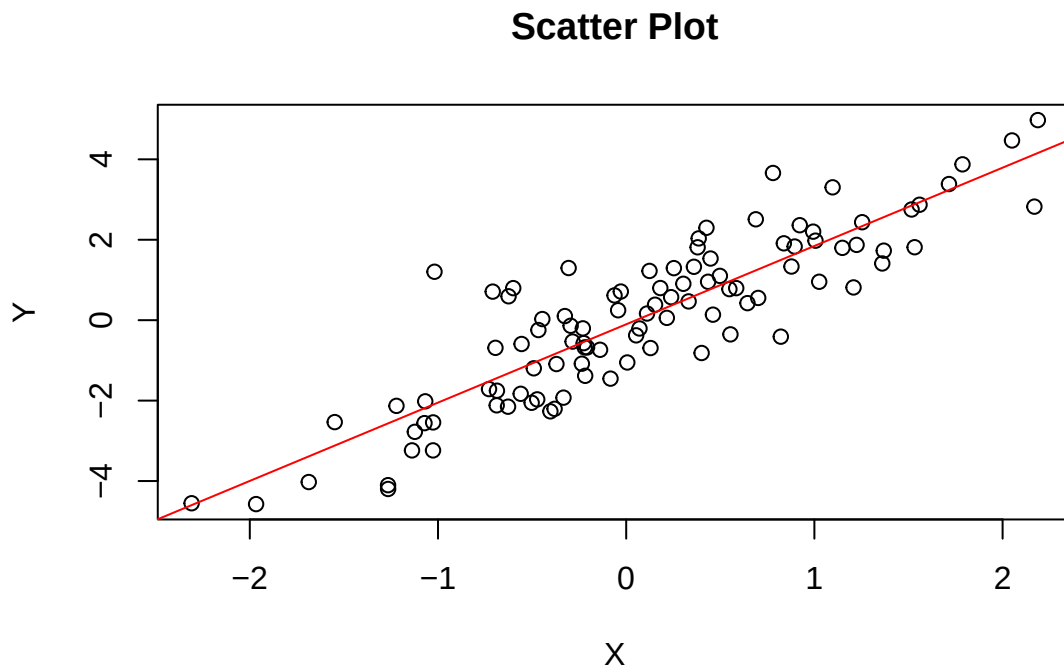
```
pip install numpy matplotlib seaborn
```

## Creating Visualisations

### R Example

```
plot(x, y, main="Scatter Plot", xlab="X", ylab="Y")
abline(lm(y ~ x), col="red")
```

Figure 1: Scatter plot of x vs y



### Python Example

# Requires matplotlib in the Python used by R (reticulate). Set eval: true locally after py\_install("matplotlib")

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(6, 4))
plt.scatter(x, y, alpha=0.6)
plt.plot(x, 2*x, color="red", linestyle='--')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Scatter Plot')
plt.show()
```

### Inline Code

Embed code results directly in your text:

#### R in Quarto:

- Markdown Syntax: There are `{r} length(x)` data points
  - Output: There are 100 data points
- Markdown Syntax: The correlation is `{r} round(cor(x,y), 3)`
  - Output: The correlation is 0.879

#### Python in Quarto:

- Markdown Syntax: There are `{python} len(x)` data points
  - Output: There are `{python} len(x)` data points

- **Markdown Syntax:** The correlation is `{python} f"{np.corrcoef(x, y)[0,1]:.3f}"`
  - **Output:** The correlation is `{python} f"{np.corrcoef(x, y)[0,1]:.3f}"`

#### **i** Note: Inline Python

Inline Python expressions require a Python engine to be enabled in Quarto. In some environments, the Python results above may not display correctly.

To enable inline Python in Quarto, the following conditions are typically required:

1. **Enable the Python engine**

Add the following to your YAML header:

```
jupyter: python3
```

However, this changes the execution engine of the document to Python (Jupyter). As a result, R code chunks may no longer run as expected without additional configuration. Therefore, inline Python is not recommended when mixing R and Python in the same document.

2. **Quarto version requirement**

Inline Python requires Quarto v1.4 or later. You can check your installed version by running the following in the RStudio Console:

```
system("quarto --version")
```

We recommend installing a version slightly above v1.4. Using the latest version is not necessary and may cause issues when rendering .qmd files.

3. **Updating Quarto**

Even after installing a newer version of Quarto, RStudio may continue to use its bundled version unless the system PATH is updated. You may need to install Quarto into the RStudio Quarto directory, overwriting the bundled version used by RStudio. For example:

```
C:/Program Files/RStudio/resources/app/bin/quarto
```

## Output Formats

Quarto supports multiple output formats from the same .qmd file:

### **HTML** (html)

- Interactive elements
- Responsive design
- Easy to share online

### **PDF** (pdf)

- Professional appearance
- Suitable for printing
- Requires LaTeX

## **Word (docx)**

- Easy to edit further
- Familiar to many users
- Suitable for collaboration

## **Presentations**

- RevealJS (format: revealjs) — HTML slides
- Beamer (format: beamer) — PDF slides
- PowerPoint (format: pptx) — PowerPoint slides

## **Best Practices**

### **Reproducibility**

- Avoid hard-coding results in your text
- Use inline code for numbers and statistics
- Set random seeds for reproducible results
- Document your data sources

### **Organisation**

- Use meaningful chunk labels
- Group related code chunks together
- Keep chunks focused on single tasks
- Use comments to explain complex code

### **Version Control**

- Track your .qmd files in Git
- Do not track generated output files
- Use .gitignore to exclude temporary files

## **Task: Create Your First Report**

1. Create a new Quarto document (.qmd)
2. Add a YAML header with your information
3. Write a brief introduction about reproducible reporting
4. Create a code chunk that generates some data
5. Create a visualisation
6. Add inline code to report statistics
7. Render your document to HTML and PDF

## Resources

- Quarto: [Quarto Documentation](#)
- Markdown: [Markdown Guide](#)
- R Markdown (optional reference): [R Markdown: The Definitive Guide](#)
- Jupyter (optional reference): [Jupyter Documentation](#)

## Quarto Introduction

### Three Components

Quarto documents rely on three main components:

1. **Markdown** for formatted text
2. **Code chunks** for executable code (R, Python, etc.)
3. **YAML** for document metadata and output formats

### Markdown for Formatted Text

Markdown is a set of conventions for formatting plain text. You can use Markdown to indicate:

- **Bold** and *italic* text: Surround italic text with asterisks, like this *without realising it*. Surround bold text with two asterisks, like this **easy to use**.

💡 Examples:

| Markdown Syntax      | Output             |
|----------------------|--------------------|
| <b>**bold text**</b> | <b>bold text</b>   |
| <i>*italic text*</i> | <i>italic text</i> |

- Lists: Group lines into bullet points that begin with asterisks or hyphens. Leave a blank line before the first bullet.

💡 Examples:

| Markdown Syntax | Output         |
|-----------------|----------------|
| - Item 1        | • Item 1       |
| - Sub-item 1    | - Sub-item 1   |
| - Sub-item 2    | - Sub-item 2   |
| - Item 2        | • Item 2       |
| 1. First item   | 1. First item  |
| 2. Second item  | 2. Second item |

- Headers (e.g., section titles): Place one or more hashtags at the start of a line to create a header (or sub-header). For example, # Say Hello to Markdown. A single hashtag (#) creates a first-level header, two hashtags (##) create a second-level header, and so on.
- Hyperlinks: Create clickable links using [text] (url).

#### 💡 Examples:

| Markdown Syntax                              | Output                                                 |
|----------------------------------------------|--------------------------------------------------------|
| [GitHub] (https://github.com)                | <a href="https://github.com">GitHub</a>                |
| [Stack Overflow] (https://stackoverflow.com) | <a href="https://stackoverflow.com">Stack Overflow</a> |
| [Reddit] (https://www.reddit.com)            | <a href="https://www.reddit.com">Reddit</a>            |

- and much more

The conventions of Markdown are simple and unobtrusive, making Markdown documents easy to read and write.

You can access a quick guide in RStudio via → “Help” → “Markdown Quick Reference”.

For more details, see the Quarto guide on [Markdown Basics](#).

## Chunks

Now, you are familiar with Markdown and Quarto. Quarto supports **code chunks** that allow you to include executable R or Python code directly in your document, making it easy to create reproducible reports.

### Code Chunks

You can use **CTRL + ALT + I** (Windows) or **OPTION + COMMAND + I** (Mac) to insert a new code chunk.

It is good practice to assign a label to each chunk using #| label:, for example #| label: simulate-data. Labels are optional but must be unique if used. This helps with debugging and cross-referencing.

### Chunk Options

In Quarto, chunk options are specified using #| inside the code block.

- echo: false: hide code but show results
- results: "hide": show code but hide results
- include: false: run code but show neither code nor output
- warning: false: suppress warnings
- message: false: suppress messages

```
set.seed(123)
x <- rnorm(100) # 100 random variables from a standard normal distribution
y <- 2 * x + rnorm(100)
cor(x, y)
```

## YAML for Render Parameters

```
title: "your awesome title"
author: "your name"
date: "today" or specify any date
```

In Quarto, the format field determines the output format when you render a document.

The format field supports the following values:

- `html`: creates HTML output (default)
- `pdf`: creates PDF output
- `docx`: creates Word output

You can also render your document as a slideshow:

- `revealjs`: creates an HTML slideshow
- `beamer`: creates a PDF slideshow
- `pptx`: creates a PowerPoint presentation

Unlike R Markdown, Quarto provides built-in support for cross-referencing, so no additional output format (e.g., bookdown) is required.

Below are examples of format options (each can be used in a complete YAML header).

### 💡 Examples of Format Options

#### Table of Contents (TOC)

```
title: "My Report"
format:
  html:
    toc: true
    toc-depth: 3
```

- `toc: true`: enables the table of contents
- `toc-depth: 3`: controls how many header levels are included

#### Section Numbering

```
title: "My Report"
format:
  pdf:
    number-sections: true
```

- `number-sections: true`: adds numbering to section headers

#### Figure Caption Placement

```
title: "My Report"
format:
  html:
    fig-cap-location: bottom
```

- `fig-cap-location: bottom`: controls where figure captions appear (top, bottom, or margin)

### Code Folding (HTML only)

```
title: "My Report"
format:
  html:
    code-fold: true
```

- `true`: folds all code by default (click to expand)
- `false`: shows all code (default)
- `show`: enables folding but keeps code expanded initially

Code folding is only available in HTML output. It does not apply to PDF or Word formats.

**Some options may not apply to all formats. More options for each format are available at: [HTML](#), [PDF](#) and [Revealjs](#).**

### Rendering

To render your `.qmd` file into HTML, PDF, or Word output, click the “**Render**” button in RStudio.

- A drop-down menu allows you to select the output format (if multiple formats are specified in the YAML header). You can try rendering to different formats.
- A shortcut on Windows is **CTRL + SHIFT + K**, and on macOS it is **SHIFT + COMMAND + K**.

When you render the document, Quarto will process your `.qmd` file and generate the output based on the YAML settings and Markdown content. The output file will be saved in your working directory, and a preview will be displayed in RStudio.

## Advanced Features in Quarto

### Figures

You can control figure appearance using chunk options:

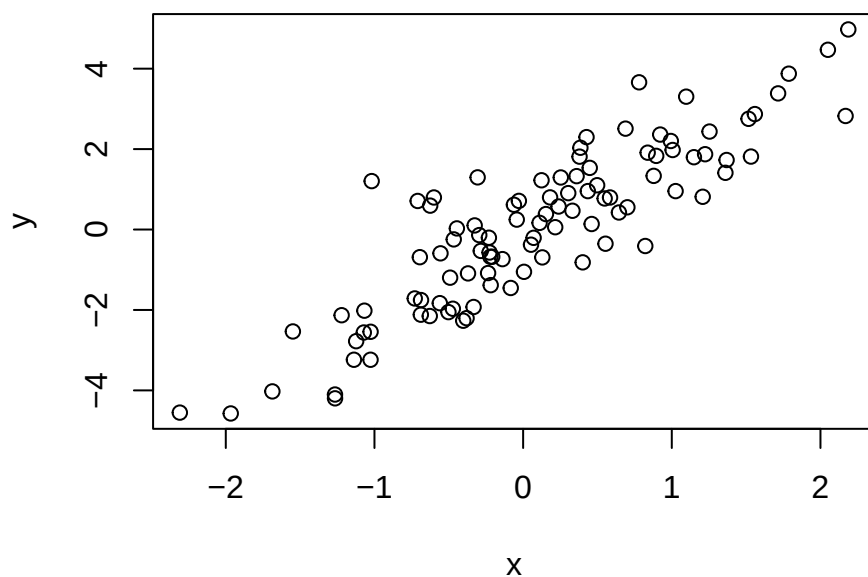
```
```{r}
#| label: fig-scatterplotfig
#| echo: true
#| fig-width: 5
#| fig-height: 4
#| fig-cap: "Scatterplot"

plot(x,y)
```
```

Note that:

- `include: false` — hides all code, results, and figures
- `results: "hide"` — hides textual output but keeps figures
- `fig-show: "hide"` — hides figures

Figure 2: Scatterplot



### Including External Images

You can include an external image using Markdown:

```
![[Caption for the picture.](unsw.jpg)] .
```

A more flexible approach is to use a code chunk:

```
```{r}
#| label: unsw
#| echo: true
#| fig-align: center
#| out-width: "50%"
#| fig-cap: "Welcome to UNSW"

knitr::include_graphics('unsw.jpg')
```
```

Figure 3: Welcome to UNSW



This allows you to control alignment, size, captions, and cross-referencing.

## Global Execution and Chunk Options

You may find yourself using similar chunk options throughout a document. Instead of repeating them in every chunk, you can define global defaults at the beginning of the document.

There are two common ways to set global options in Quarto:

1. Use the YAML header to set document-level options
2. Use `knitr::opts_chunk$set()` for R chunks when using the knitr engine

### Global Options in the YAML Header

Quarto allows many execution and output options to be set globally in the YAML header. These options apply to the document unless overridden in an individual code chunk.

For example:

```
---
title: "My Report"
format:
  html:
    code-fold: true
    fig-width: 6
    fig-height: 4
execute:
  echo: true
  warning: false
  message: false
```

Here: - `execute: echo: true` shows code by default - `execute: warning: false` suppresses warnings by default - `execute: message: false` suppresses messages by default - `fig-width` and `fig-height` set the default figure size for the HTML output - `code-fold: true` enables code folding in HTML output

You can still override these options in a specific chunk. For example:

```
```{r}
#| echo: false

summary(cars)
```
```

In this chunk, `echo: false` overrides the global `echo: true`.

#### Note

For more details on available execution and output options, see the Quarto guide on [Execution Options](#).

### Global Options for R Chunks Using `knitr`

For R chunks, you can also set global defaults using `knitr::opts_chunk$set()` in an initial setup chunk:

These options apply to subsequent R chunks unless overridden locally.

For example, you can override the figure height in a specific R chunk:

```

```{r}
#| label: taller-figure
#| echo: true
#| fig-height: 6
#| fig-cap: "Multiple plots"

par(mfrow=c(2,2))
for(i in 1:4){
  plot(x[(1:25)*i], y[(1:25)*i])
}
```

```

### **i** Note

For Quarto documents that mix R and Python, YAML-level options are usually clearer and more consistent. `knitr::opts_chunk$set()` is mainly for R chunks using the knitr engine and should not be relied on to control Python chunks.

## Cross-Referencing

Quarto provides built-in support for cross-referencing figures, tables, equations, and sections.

To create a cross-reference, you need to assign a **label** to an element. A label is a unique identifier that allows the element to be referenced elsewhere in the document.

Each label uses a **prefix** to indicate the type of element. Common prefixes include:

- `sec-` for sections
- `fig-` for figures
- `tbl-` for tables
- `eq-` for equations

Once a label is defined, you can refer to it in the text, and Quarto will automatically format the reference and assign numbers where appropriate (e.g., “Section 2”, “Figure 1”, etc.).

### **i** Naming tips

- Labels must be **unique** within the document
- Use **kebab-case** (e.g., `sec-my-section`)
- Avoid using underscores (`_`) in labels

## Figures and Tables

We can refer to figures and tables using their labels.

For example, the following figure was defined earlier:

- Markdown Syntax: `@fig-scatterplotfig`

- Output: see Figure 2

### Table example

To create and reference a table:

```
```{r}
#| label: tbl-mtcars
#| tbl-cap: "First few rows of mtcars dataset"

knitr::kable(head(mtcars))
```
```

Example:

```
knitr::kable(head(mtcars))
```

Table 4: First few rows of mtcars dataset

|                   | mpg  | cyl | disp | hp  | drat | wt    | qsec  | vs | am | gear | carb |
|-------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| Mazda RX4         | 21.0 | 6   | 160  | 110 | 3.90 | 2.620 | 16.46 | 0  | 1  | 4    | 4    |
| Mazda RX4 Wag     | 21.0 | 6   | 160  | 110 | 3.90 | 2.875 | 17.02 | 0  | 1  | 4    | 4    |
| Datsun 710        | 22.8 | 4   | 108  | 93  | 3.85 | 2.320 | 18.61 | 1  | 1  | 4    | 1    |
| Hornet 4 Drive    | 21.4 | 6   | 258  | 110 | 3.08 | 3.215 | 19.44 | 1  | 0  | 3    | 1    |
| Hornet Sportabout | 18.7 | 8   | 360  | 175 | 3.15 | 3.440 | 17.02 | 0  | 0  | 3    | 2    |
| Valiant           | 18.1 | 6   | 225  | 105 | 2.76 | 3.460 | 20.22 | 1  | 0  | 3    | 1    |

- Markdown Syntax: @tbl-mtcars
- Output: see Table 4

### Equations

You can write equations using LaTeX syntax:

```
$$
\bar{X} = \frac{\sum_{i=1}^n X_i}{n}
$$ {#eq-mean}
```

Example:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \quad (1)$$

- Markdown Syntax: @eq-mean
- Output: see Equation 1

## Sections

You can assign a label to a section by adding `{#label}` to the header:

```
### Sections {#sec-sections}
```

You can then refer to this section in text:

- Markdown Syntax: `@sec-sections`
- Output: see Section

## Citations

To include references in a Quarto document:

1. **Create a bibliography file:** Create a `.bib` file in the same directory as your `.qmd` file (e.g., `refWeek1.bib`).
2. **Add references to the file:** You can obtain BibTeX entries from sources such as [Google Scholar](#). For example:

- Search for a paper (e.g., *Handbook of Data Visualization*)
- Click **Cite** → **BibTeX**
- Copy the entry and paste it into your `.bib` file

3. **Specify the bibliography in YAML:** Add the following to your YAML header:

```
bibliography: refWeek1.bib
```

4. **Cite in text:** Use `@key` to cite a reference, where `key` is the identifier in the `.bib` file. For example:

- Markdown Syntax: `@chen2007handbook`
- Output: Chen, Härdle, and Unwin ([2007](#))

5. **Add a reference section:** At the end of your document, include:

```
# References
```

Quarto will automatically generate the reference list.

## Comments

You can add comments in your document that will not appear in the final output.

```
<!-- This text will not be displayed in the output -->
```

Example (no text will appear below because it is commented out):

You can also use the RStudio menu:

click **Code** → **Comment/Uncomment Lines** to quickly comment or uncomment selected lines.

## Visual Markdown Editor in RStudio

The visual editor is useful for those who are not yet familiar with Markdown or prefer not to write Markdown syntax manually. In RStudio, you can switch between **Source** mode and **Visual** mode for a .qmd document by clicking the “Visual” or “Source” button in the editor toolbar.<sup>1</sup>

As an exercise, you can experiment with the following features of the visual editor:

- Format: **bold**, *italic*, underline, code, super<sup>script</sup>, sub<sub>script</sub>, etc.
- Headers: 1st-level header, 2nd-level header, . . . , 6th-level header
- Hyperlinks
- Lists (bulleted or numbered)
- Insert: citations, cross-references (figures, tables, equations), footnotes, and comments
- And so on

In general, the visual editor makes document editing feel more like working in Word. However, for reproducible documents, it is still important to understand the underlying Markdown and Quarto syntax. Be careful when editing YAML headers, code chunks, and LaTeX equations, as visual editing may occasionally change the underlying source in unexpected ways.

## References

Chen, Chun-houh, Wolfgang Karl Härdle, and Antony Unwin. 2007. *Handbook of Data Visualization*. Springer Science & Business Media.

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<sup>1</sup>If you do not see this option, please update RStudio to a recent version.